Convex blocks and a coarse Dobrushin chain: a Poincaré bound for the core of the double-well path measure with constants polynomial in $\hbar$

Yoshikazu Hosomura
with the assistance of Claude (Claude Code, Anthropic)

2026-10-17 (version 2; version 1: 2026-10-15)

wvbks0 [at] gmail.com · https://computoergosum.com/en/index.html

AI use is disclosed in §10; what is not claimed is in §9.

Abstract

We consider the Euclidean path measure of the quartic double well $V(z) = (z^2-1)^2/8$ on a time lattice of spacing $\varepsilon = 1/2$, restricted to the box $[-x_c, 0]^k$ with $x_c = 0.89$ and with both outer boundary values in the well, $\alpha, \beta \le -x_c$; this "core" measure carries the weight $e^{-S/\hbar}$ and the Dirichlet form $(\hbar/\varepsilon)\sum_i |\partial_i f|^2$. We prove three Poincaré bounds on paper. Theorem B: a block-heat-bath path coupling and Ollivier's contraction-to-gap criterion turn a block Poincaré constant C_B and a Dobrushin–Shlosman type average influence $\delta < 1$ into $C_P \le C_B/(1-\delta)$ for every k. Theorem BP: for blocks of length at most 15 and boundary values in $(-\infty, 0]$, $C_B \le 9.18 + c_1\hbar^{-8}$ with $\ln c_1 \le 89.38$, for all $\hbar \in (0, 1]$; the exponential $e^{c/\hbar}$ is removed by a hidden convexity of the action in the variables z_i^2 and a one-dimensional Hardy-type bound. Theorem CC: for the coarse chain $y_m = z_{8m}$ with Dobrushin column sum $\rho_{\mathrm{col}}(\hbar) < 1$, $C_P \le 9.18 + 455.84\, C_{\mathrm{site}}(\hbar)/(1-\rho_{\mathrm{col}}(\hbar))$ with $C_{\mathrm{site}} \le e^{80.79}\hbar^{-8}$, uniformly in k. Theorems BP and CC use, in addition, the Bakry–Émery criterion on convex domains and the Brascamp–Lieb variance inequality, which are standard results and are cited. The hypothesis $\rho_{\mathrm{col}} < 1$ is verified on paper for $\hbar \in [0.27, 1]$ up to one one-dimensional integral evaluated in floating point; for every $\hbar \in [10^{-5}, 1]$ by an interval-arithmetic certificate over 239 intervals of $\hbar$ that cover $[10^{-5}, 1]$ without gaps and over every geometry of the coarse chain, giving $\rho_{\mathrm{col}} \le 0.400$ (version 2; version 1 had 23 values of $\hbar$ in double precision with outward margins); and for $0 < \hbar \le 10^{-5}$ on paper together with one one-dimensional interval computation (Proposition PIN), giving $\rho_{\mathrm{col}} \le 0.874$. Hence $C_P \le 9.18 + e^{88.99}\hbar^{-8}$ for all $\hbar \in (0, 1]$ and all k. The algebraic steps are written in Lean 4; the analysis is on paper. The model is one-dimensional; nothing is claimed about four-dimensional lattice gauge theory.

1 Introduction

A Poincaré inequality for a Gibbs measure in a double-well potential typically has a constant of order $e^{\Delta/\hbar}$, with Δ the barrier height (Kramers). Restricting to a single well removes the barrier, but the restricted measure is not log-concave: the potential V has $V'' < 0$ on $(-1/\sqrt{3}, 0]$ and the box reaches up to the top of the barrier at 0. The standard routes from a Dobrushin–Shlosman condition to a Poincaré or log-Sobolev inequality bound block constants by the Holley–Stroock perturbation lemma [HS] and therefore pay a factor exponential in the interaction strength $1/(\varepsilon\hbar)$ (for instance the block constant $c_0 e^{4|X|\|\Phi\|}$ in [SZ]; see also [Ma]). The aim of this paper is a Poincaré constant for the single-well core that is *polynomial* in $\hbar$ and uniform in the length k.

The question arises in an online textbook by the authors [KSR, Part 10], where a Poincaré bound of this form for the core is one of the rows on which a conditional argument for the double-well path measure rests (the row labelled ($\mathrm{T0}'_P$) there). A companion note [KSR2] proves the opposite direction: an inverse squared spectral gap bounds the Langevin Poincaré constant of Euclidean path measures *from below*. The present paper concerns the *upper* bound for the core, which is a different object; the two results are not used in each other's proofs.

What the paper does.

- Theorem 3.1 (§3) is an abstract step: a path coupling for the block heat bath, combined with Ollivier's criterion [Ol, Prop. 29, Cor. 30], gives $C_P \le C_B/(1-\delta)$. No step introduces $e^{1/(\varepsilon\hbar)}$.
- Theorem 4.5 (§4) bounds the block constant polynomially in $\hbar$.
- Theorem 5.3 (§5) replaces the block chain by a one-site Dobrushin chain on every eighth site. It needs neither Theorem 3.1 nor Theorem 4.5 as stated, only their ingredients, and its single numerical hypothesis is $\rho_{\mathrm{col}}(\hbar) < 1$.
- Proposition 6.4 (§6) states what is known about $\rho_{\mathrm{col}}(\hbar) < 1$, with the strength of each part named in the statement.

2 Setting

Fix $\varepsilon = 1/2$, $V(z) = (z^2-1)^2/8$, $x_c = 0.89$ and $\hbar \in (0,1]$. For $k \ge 1$ and outer boundary values $\alpha, \beta \in \mathbb{R}$ put $z_0 = \alpha$, $z_{k+1} = \beta$ and

$$S(z) = \sum_{i=0}^{k} \frac{(z_{i+1}-z_i)^2}{2\varepsilon} + \sum_{i=1}^{k} \varepsilon V(z_i), \qquad \mu_k^{\alpha,\beta,\hbar}(dz) = \frac{1}{Z}\, e^{-S(z)/\hbar}\, \mathbf{1}_{\Omega_k}(z)\, dz, \quad \Omega_k = [-x_c, 0]^k.$$

The Dirichlet form is $\mathcal{E}(f) = (\hbar/\varepsilon) \sum_{i=1}^{k} \mu |\partial_i f|^2$ (Neumann conditions on $\partial\Omega_k$), and $C_P(\mu)$ is the least constant with $\mathrm{Var}_\mu f \le C_P(\mu)\, \mathcal{E}(f)$ for Lipschitz f. The *core* is the family $Q_k = \mu_k^{\alpha,\beta,\hbar}$ with

$$\textbf{(END)} \qquad \alpha, \beta \le -x_c,$$

that is, both outer neighbours sit in the well. (For $a > 0$ the boundary term is not compatible with the hidden convexity below; the restriction (END) is part of every statement.)

The Hessian of S in the z coordinates is $\varepsilon^{-1} T_n + \varepsilon \operatorname{diag} V''(z_i)$ with $T_n = \mathrm{tridiag}(-1, 2, -1)$, and $V'' \ge V''(0) = -1/2$. Hence on any block of n consecutive free sites, for arbitrary real boundary values,

$$\operatorname{Hess} S \;\ge\; \kappa(n)\, I, \qquad \kappa(n) := \tfrac{15}{4} - 4\cos\!\big(\tfrac{\pi}{n+1}\big). \tag{1}$$

Numerically $\kappa(6) = 0.1461$, $\kappa_0 := \kappa(7) = 0.054482$ and $\kappa(8) = -0.00877$. Blocks of length ≤ 7 are uniformly convex in z; length 8 is the first that is not. Every construction below cuts the chain into pieces of length ≤ 7.

Hidden convexity ($\mathbf{K}'$). For $z \le 0$ put $\rho_i = z_i^2$. Then $(z_i - z_j)^2 = \rho_i + \rho_j - 2\sqrt{\rho_i \rho_j}$ is jointly convex in (ρ_i, ρ_j) (the geometric mean is concave) and $\varepsilon V(-\sqrt{\rho}) = \varepsilon(\rho-1)^2/8$ is convex. Hence, for boundary values $a, b \le 0$, $\tilde{S}(\rho) := S(-\sqrt{\rho})$ is convex on $[0, x_c^2]^n$ (indeed $(\varepsilon/4)$-strongly convex), jointly in (ρ, a^2, b^2), and its minimiser is unique. This is proved in [KSR, Part 10] and its algebraic core is in Lean (§7). It is used below only through partial minimisation: if $\tilde{S}$ is jointly convex then $\sigma \mapsto \min_{\rho'} \tilde{S}(\rho', \sigma)$ is convex.

The coarse chain. For $k \geq 8$ let $M = \lfloor k/8 \rfloor$, $y_m := z_{8m}$ $(1 \leq m \leq M)$, and let $B_0 = \{1, \dots, 7\}$, $B_m = \{8m+1, \dots, \min(8m+7, k)\}$ be the blocks between coarse sites. With the coarse sites frozen, the blocks are independent (Markov property) and each is κ_0-convex by (1). The law ν of y is a nearest-neighbour chain on $[-x_c, 0]^M$ with effective pair interaction $w_B(a,b) = -\hbar \ln \int e^{-S_B(z;a,b)/\hbar} dz$. The conditional law $\nu(dy_m \mid y_{m-1} = a,\, y_{m+1} = b)$ is the marginal of the middle site z_8 of a chain $z_1, \dots, z_N$, $N = 8 + r$, with boundary values $z_0 = a \in [-x_c, 0]$ and $z_{N+1} = b$; in the interior $r = 7$ and $b \in [-x_c, 0]$, at the right end of the chain $r \in \{0, \dots, 7\}$ and $b = \beta \leq -x_c$. Its *influence* (a Dobrushin coefficient [Do]) is

$$\rho(a,b;r) := \partial_a \mathbb{E}[z_8 \mid a, b] = \frac{\mathrm{Cov}(z_1, z_8)}{\varepsilon\hbar},$$
$$\rho_{\mathrm{col}}(\hbar) := \sup_m \sum_{m'=m\pm1} \sup_{a,b} \rho_{m' \leftarrow m} \;\leq\; 2 \sup_{r,a,b} \rho(a,b;r).$$

What is shown and what is not. Theorems 3.1, 4.5 and 5.3 are proved on paper, with cited ingredients listed in Table 2. Theorem 5.3 is conditional on $\rho_{\mathrm{col}}(\hbar) < 1$. That hypothesis is established for every $\hbar \in (0, 1]$: on $[10^{-5}, 1]$ by the interval-arithmetic certificate of Proposition 6.4(b), under the assumptions of §6.3, and on $(0, 10^{-5}]$ by Proposition 6.6, on paper with one one-dimensional interval computation. Accordingly, for the core in which both outer values lie in the well (the compartment-side core of [KSR, Part 10]), the row $(\mathrm{T0}'_P)$ moves from an assumption to a statement that is conditional on the cited ingredients and on the stated assumptions of the certificate (Table 2); it does not become unconditional. The box-side core of that argument (its Proposition H4), in which an outer neighbour lies beyond the summit and the conditional law is bistable with constants of type $e^{\Delta S/\hbar}$ [KSR, 10-11 §06], is not covered by the hypothesis $\alpha, \beta \leq -x_c$ and is not addressed here. The four-dimensional lattice gauge problem that motivates [KSR] is not addressed (§9).

3 Theorem B: block heat bath and contraction

Fix a block length L and put $B_j = [j, j+L-1] \cap [1, k]$ for $j = 2-L, \dots, k$, so that there are $N = k + L - 1$ blocks, truncated blocks included, and every site lies in exactly L of them. Let E_B be the conditional expectation given the coordinates outside B, and $P = N^{-1} \sum_j E_{B_j}$ the block heat bath. For a site y and a block B having y as a boundary site, let $J_B(y) = \sup_s \sum_{i \in B} \partial_s \mathbb{E}_B^{s,\cdot}[z_i]$, the supremum running also over the other boundary value of B, and

$$\delta := \sup_y \frac{J_{\mathrm{right}}(y) + J_{\mathrm{left}}(y)}{L}.$$

Theorem 3.1 (Theorem B)**.** *Let $\alpha, \beta \leq -x_c$ and $\hbar > 0$. Assume* (BP) *every block B_j (length $\leq L$, all boundary values in $(-\infty, 0]$) satisfies $\mathrm{Var}_{E_B}(f) \leq C_B(\hbar/\varepsilon) \sum_{i \in B} E_B |\partial_i f|^2$, and* (DS) *$\delta < 1$. Then for every $k \geq 1$,*

$$C_P(\mu_k^{\alpha,\beta,\hbar}) \leq \frac{C_B}{1-\delta}.$$

The proof uses the FKG lattice condition and Holley's criterion for monotone coupling [Ho], [Pr] (standard results, cited) and Ollivier's criterion [Ol] (checked against the arXiv version).

Proof. Monotone coupling. The cross term of $-S/\hbar$ is $+z_i z_{i+1}/(\varepsilon\hbar)$, supermodular, and the reference measure is a product of intervals; the block measures satisfy the FKG lattice condition and the density ratio between boundary values $s < s'$ is $\exp((s'-s) z_{\mathrm{first}}/(\varepsilon\hbar))$, increasing. By Holley's criterion and Strassen's theorem there is a coupling with $z \leq z'$ coordinatewise, and then $W_1(\mu_B^s, \mu_B^{s'}) = \sum_i (\mathbb{E}^{s'} z_i - \mathbb{E}^s z_i) \leq |s' - s|\, J_B(y)$.

Path coupling. If z, z' differ only at the site y, by d, then one step of P chooses a block containing y (L choices; the two configurations are coupled to agree), a block having y as a boundary site (two choices; the discrepancy is at most $d + J_B(y)d$), or another block (discrepancy d). Hence $W_1(P(z,\cdot), P(z',\cdot)) \le (1-\kappa)d$ with $\kappa = (L/N)(1-\delta)$; for general z, z' change one coordinate at a time (the box is a product) and use the triangle inequality.

Contraction to gap. P is self-adjoint and positive in $L^2(\mu)$. Since Ω_k is compact, $\|P^n f - \mu f\|_2 \le \operatorname{diam}(\Omega_k)(1-\kappa)^n \|f\|_{\mathrm{Lip}}$ for Lipschitz f, so the spectral measure of $f - \mu f$ has no mass in $(1-\kappa, 1]$; by density of Lipschitz functions, $\langle f, (I-P)f\rangle \ge \kappa \operatorname{Var} f$ for all f [Ol, Prop. 29, Cor. 30].

Comparison. $\langle f, (I-P)f\rangle = N^{-1}\sum_j \mu(\operatorname{Var}_{B_j} f) \le (C_B L/N)\,\mathcal{E}(f)$ by (BP), since each site lies in L blocks. Combining, $\operatorname{Var} f \le C_B(1-\delta)^{-1}\mathcal{E}(f)$. □

Remark 3.2. The ingredients are of known types (path coupling under a Dobrushin–Shlosman type condition [Ma], comparison of block dynamics, Ollivier's criterion). We do not know whether this combination appears in the literature in this form; in the references we were able to read, the block constant is controlled by a Holley–Stroock perturbation, which is where $e^{c/(\varepsilon\hbar)}$ enters. Here that step is the hypothesis (BP), and Theorem 4.5 supplies it. Truncated blocks enter the supremum in δ.

The algebraic steps (the one-step contraction inequality, the passage from spectral weight to gap, and the final comparison) are in Lean (§7). The quantity δ itself is not bounded on paper in this paper; Theorem 5.3 below avoids it.

4 Theorem BP: block constants polynomial in $\hbar$

For a block of length $L \le 15$ with boundary values $a, b \in (-\infty, 0]$ write $\mu_{a,b} \propto e^{-S/\hbar}\mathbf{1}_{[-x_c,0]^L}$ and $C_B(\hbar) := \sup_{L\le 15,\, a,b\le 0} C_P(\mu_{a,b})$.

Lemma 4.1 (short blocks; BE_7)**.** *If $n \le 7$, then for every $\hbar > 0$ and all real boundary values, $C_P(\mu_{a,b}) \le \varepsilon/\kappa(n) \le \varepsilon/\kappa_0 = 9.1774$.*

Proof. By (1) the potential $S/\hbar$ is $(\kappa(n)/\hbar)$-strongly convex on the convex set $[-x_c, 0]^n$. The Bakry–Émery criterion for a strongly log-concave measure restricted to a convex set (Neumann boundary) gives $\operatorname{Var} f \le (\hbar/\kappa(n)) \int |\nabla f|^2 d\mu$ [BGL], [BL, Thm. 4.1] (the criterion is standard and is cited; its application to the box $[-x_c, 0]^n$ through (1) is the step of this paper). □

Lemma 4.2 (variance split at the middle site)**.** *Let $8 \le L \le 15$, $j = \lceil L/2 \rceil$, let m be the law of z_j and C_m its Poincaré constant for $\operatorname{Var}_m g \le C_m \int |g'|^2 dm$. Then $C_P(\mu_{a,b}) \le \varepsilon/\kappa_0 + (4/(\varepsilon\kappa_0^2))\, C_m/\hbar$, with $4/(\varepsilon\kappa_0^2) = 2695.2$.*

Proof. Both sides of j have length ≤ 7. By the law of total variance, $\operatorname{Var}_\mu f = m(\operatorname{Var}_{\mu_s} f) + \operatorname{Var}_m g$, $g(s) = \mu_s(f)$, where μ_s is the law with $z_j = s$ frozen, a product of two κ_0-convex blocks; Lemma 4.1 and tensorisation give the first term $\le (\hbar/\kappa_0)A$, $A = \sum_{i\ne j} \mu|\partial_i f|^2$. Differentiating, $g'(s) = \mu_s(\partial_j f) + \operatorname{Cov}_{\mu_s}(f, h)/(\varepsilon\hbar)$ with $h = z_{j-1} + z_{j+1}$; Cauchy–Schwarz and $\operatorname{Var}_{\mu_s} h \le 2\hbar/\kappa_0$ give $|g'|^2 \le 2\mu_s|\partial_j f|^2 + 4A_s/(\varepsilon^2\kappa_0^2)$. Integrate against m and rescale to $\mathcal{E}$. □

Lemma 4.3 (one-dimensional Hardy-type bound; H1D)**.** *Let $m(dt) = Z^{-1}e^{-F(t)/\hbar}dt$ on $I = [-\ell, 0]$ with F continuous, and $\Delta := \sup_{y\le s\le x}\big[F(s) - \max(F(x), F(y))\big]$. Then $C_m \le 2\ell^2 e^{\Delta/\hbar}$.*

Proof. $\mathrm{Var}_m\, g = \iint_{y<x}(g(x)-g(y))^2 dm\, dm$ and $(g(x)-g(y))^2 \le \int_y^x |g'|^2 p \int_y^x p^{-1}$. By Fubini $\mathrm{Var}_m\, g \le \int |g'(t)|^2 p(t) K(t) dt$ with $K(t) = \iint_{y\le t\le x} \int_y^x p(x)p(y)/p(s)\, ds\, dx\, dy$. For $y \le s \le x$, $p(x)p(y)/p(s) \le e^{\Delta/\hbar}(p(x)+p(y))$, and $\iint_{y\le t\le x}(x-y)(p(x)+p(y)) \le 2\ell^2$. □

If F is quasi-convex ($\Delta = 0$), $C_m \le 2\ell^2$ independently of $\hbar$: the factor $e^{c/\hbar}$ can only come from a second valley of the free energy. The discrete form of Lemma 4.3 is in Lean; the Fubini step is on paper. We do not use Muckenhoupt's criterion [Mu].

Lemma 4.4 (quasi-convex free energy plus small oscillation; QC)**.** *Let $n = L-1 \le 14$, $F_\hbar(s) = -\hbar \log \int e^{-S(z',s)/\hbar} dz'$ the free energy of the middle site, $F_0(s) = \min_{z'} S(z',s)$ and $R = F_\hbar - F_0$. Then* (a) *F_0 is quasi-convex on $[-x_c, 0]$, and* (b) *$R(s) - R(s') \le \hbar \log(K_1 \hbar^{-n/2})$ for all s, s', where $K_1 = (2\pi/\kappa_0)^{n/2} G^n e^{n(1+\Lambda/(2G^2))}$, $G = 6x_c/\varepsilon + \varepsilon \max|V'| = 10.776$, $\Lambda = 4/\varepsilon + \varepsilon \max V'' = 8.344$. Consequently $\Delta(F_\hbar) \le \hbar \log(K_1 \hbar^{-n/2})$.*

Proof. (a) $F_0(s) = \tilde F_0(s^2)$ with $\tilde F_0(\sigma) = \min_{\rho'} \tilde S(\rho', \sigma)$ convex by (K′) and partial minimisation; $s \mapsto s^2$ is monotone on $[-x_c, 0]$, so the sublevel sets of F_0 are intervals. (b) Let $z^{(s)}$ be the minimiser (unique by Lemma 4.1), $g = \nabla_{z'} S(z^{(s)}, s)$ and $d = z' - z^{(s)}$. The KKT condition gives $\langle g, d\rangle \ge 0$ on the box, and $(\kappa_0/2)|d|^2 + \langle g, d\rangle \le S - F_0 \le (\Lambda/2)|d|^2 + \langle g, d\rangle$. The upper bound gives $\int e^{-(S-F_0)/\hbar} \le (2\pi\hbar/\kappa_0)^{n/2}$; on the cube obtained by moving each coordinate inward by $\hbar/G$ one has $\langle g, d\rangle \le n\hbar$, whence $\int e^{-(S-F_0)/\hbar} \ge (\hbar/G)^n e^{-n(1+\Lambda\hbar/(2G^2))}$ when $|a|, |b| \le 3x_c$; for $|a| > 3x_c$ the first coordinate of the minimiser is pinned at $-x_c$ and the same estimate holds with the first direction rescaled by $\hbar/g_1$. A quasi-convex function plus an oscillation of size D has barrier at most D. □

Theorem 4.5 (Theorem BP)**.** *For all $\hbar \in (0, 1]$, $L \le 15$ and $a, b \in (-\infty, 0]$,*

$$C_P(\mu_{a,b}) \le \frac{\varepsilon}{\kappa_0} + c_1 \hbar^{-8}, \qquad \frac{\varepsilon}{\kappa_0} = 9.1774, \quad \ln c_1 \le 89.38,$$

and $C_P(\mu_{a,b}) \le \varepsilon/\kappa(L) \le 9.18$ if $L \le 7$. The bound uses (K′) *and Lemma 4.1, whose convex-domain Bakry–Émery step is cited.*

Proof. Lemmas 4.3 and 4.4 give $C_m \le 2x_c^2 K_1 \hbar^{-n/2}$, and Lemma 4.2 gives $C_P \le \varepsilon/\kappa_0 + (8x_c^2/(\varepsilon\kappa_0^2)) K_1 \hbar^{-1-n/2}$. For $n \le 14$ and $\hbar \le 1$, $\hbar^{-1-n/2} \le \hbar^{-8}$ and K_1 is largest at $n = 14$: $\ln(8x_c^2/(\varepsilon\kappa_0^2)) = 8.359$ and $\ln K_1 = 33.234 + 33.283 + 14.503 = 81.020$, so $\ln c_1 \le 89.379$. □

The constant is crude. (K′) is used once, in Lemma 4.4(a), to exclude a second valley of $F_\hbar$; without it $\Delta = O(1)$ would be possible and the bound would be of Kramers type $e^{\Delta/\hbar}$. With Theorem 3.1, Theorem 4.5 gives $C_P(Q_k) \le (9.18 + c_1\hbar^{-8})/(1-\delta)$ whenever the averaged influence δ for $L = 15$ is < 1; we do not certify δ here and use Theorem 5.3 instead.

5 Theorem CC: the coarse Dobrushin chain

Let $M_7 = \varepsilon^{-1} T_7 - (\varepsilon/2) I$ (the worst-case Hessian of a block of length 7), so $\lambda_{\min}(M_7) = \kappa_0$, and

$$g_7 := (M_7^{-1})_{11} = (M_7^{-1})_{77} = \tfrac{1832324}{1770735} = 1.03478,$$

computed by the continued fraction of the tridiagonal Schur recursion (value in Lean). For a block B of length $n \le 7$, $(M_n^{-1})_{11} \le g_7$.

Lemma 5.1 (stochastic monotonicity of the coarse conditional laws)**.** *$\nu(\cdot \mid a, b)$ is increasing in a in the likelihood-ratio order, so that $W_1(\nu(\cdot \mid a, b), \nu(\cdot \mid a', b)) = |\mathbb{E}_{a'} y_m - \mathbb{E}_a y_m| \le \sup \rho \cdot |a' - a|$.*

Proof. The law of z_8 given a is the image of the Markov chain of Lemma 6.2 below, whose kernels are increasing in their starting point in the likelihood-ratio order (Lemma 6.3(a)); composition of TP_2 kernels is TP_2 [Ka]. Hence the law of z_8 is increasing in a, and the monotone coupling realises W_1 as the difference of means. □

Theorem 5.2 (coarse chain)**.** *If $\rho_{\mathrm{col}} < 1$, then for every Lipschitz g on $[-x_c, 0]^M$,*

$$\mathrm{Var}_\nu\, g \le \frac{C_{\mathrm{site}}}{1-\rho_{\mathrm{col}}}\,\frac{\hbar}{\varepsilon}\sum_m \nu|\partial_{y_m} g|^2, \qquad C_{\mathrm{site}} := \sup_{m,a,b}\frac{\varepsilon}{\hbar}C_P(\nu(\cdot \mid a,b)).$$

Proof. This is Theorem 3.1 with blocks of one coarse site. Let $P = M^{-1}\sum_m P_m$ be the one-site heat bath. If y, y' differ only at m, by d, choosing m couples them to agree, choosing $m' = m\pm1$ costs at most $d + \rho_{m'\leftarrow m}d$ by Lemma 5.1, and other choices leave d; so $W_1(P(y,\cdot), P(y',\cdot)) \le (1-(1-\rho_{\mathrm{col}})/M)d$. Ollivier's criterion gives $\langle g, (I-P)g\rangle \ge ((1-\rho_{\mathrm{col}})/M)\,\mathrm{Var}_\nu\, g$, and $\langle g,(I-P)g\rangle = M^{-1}\sum_m \nu(\mathrm{Var}_m\, g) \le M^{-1}C_{\mathrm{site}}(\hbar/\varepsilon)\sum_m \nu|\partial_m g|^2$. □

The conditional law $\nu(\cdot \mid a,b)$ is the marginal of the middle site of a chain of at most 15 sites whose two halves have length ≤ 7. Lemmas 4.3 and 4.4 (with (K′) extended to the boundary values, §2) therefore give

$$C_{\mathrm{site}} \le 2\varepsilon x_c^2 K_1(14)\,\hbar^{-8} = e^{80.79}\hbar^{-8}. \tag{2}$$

Theorem 5.3 (Theorem CC)**.** *Let $\varepsilon = 1/2$, $V(z) = (z^2-1)^2/8$, $x_c = 0.89$, $\hbar \in (0,1]$ and $\alpha, \beta \le -x_c$. Assume* (DOB) *$\rho_{\mathrm{col}}(\hbar) < 1$. Assume moreover the cited statements* BE_7 *(Lemma 4.1) and* BL*: for a measure on a convex set with* $\mathrm{Hess}(S/\hbar) \ge M/\hbar$, $\mathrm{Var}\langle e,z\rangle \le \hbar\langle e, M^{-1}e\rangle$ *[BL, Thm. 4.1] (a standard result, cited). Then for every $k \ge 1$,*

$$C_P(Q_k) \le \frac{\varepsilon}{\kappa_0} + c_6\,\frac{C_{\mathrm{site}}(\hbar)}{1-\rho_{\mathrm{col}}(\hbar)} \le 9.18 + \frac{455.84\,e^{80.79}\,\hbar^{-8}}{1-\rho_{\mathrm{col}}(\hbar)}, \qquad c_6 := \max\Big(3, \frac{6g_7}{\varepsilon^2\kappa_0}\Big) = 455.84.$$

In particular, if $\rho_{\mathrm{col}}(\hbar) \le 1/2$ then $C_P(Q_k) \le 9.18 + e^{87.61}\hbar^{-8}$, uniformly in k.

Proof. For $k \le 7$ the core is one block and Lemma 4.1 gives $C_P \le 9.18$. Let $k \ge 8$, write $A = \sum_{i \text{ not coarse}} \mu|\partial_i f|^2$ and $B = \sum_m \mu|\partial_{8m} f|^2$, so $\mathcal{E}(f) = (\hbar/\varepsilon)(A+B)$.

1. $\mathrm{Var}\, f = \nu(\mathrm{Var}(f \mid y)) + \mathrm{Var}_\nu\, g$, $g(y) = \mathbb{E}[f \mid y]$.
2. Given y the blocks are independent and κ_0-convex; by Lemma 4.1 and tensorisation, $\nu(\mathrm{Var}(f \mid y)) \le (\varepsilon/\kappa_0)(\hbar/\varepsilon)A$.
3. $\partial_{y_m} g = \mathbb{E}[\partial_{8m} f \mid y] + [\mathrm{Cov}(f, z_{8m-1} \mid y) + \mathrm{Cov}(f, z_{8m+1} \mid y)]/(\varepsilon\hbar)$, since $\partial_{y_m} S = (2y_m - z_{8m-1} - z_{8m+1})/\varepsilon + \varepsilon V'(y_m)$ and terms depending on y_m alone do not contribute to covariances.
4. $X = z_{8m\pm1}$ depends on one block B only, so $\mathrm{Cov}(f, X \mid y) = \mathrm{Cov}(f_B, X \mid y)$ with $f_B = \mathbb{E}[f \mid y, x_B]$. Then $\mathrm{Var}(f_B \mid y) \le (\hbar/\kappa_0)A_B$ (block Poincaré and Jensen) and $\mathrm{Var}(X \mid y) \le \hbar g_7$ (BL), so $(\mathrm{Cov}/(\varepsilon\hbar))^2 \le g_7 A_B/(\varepsilon^2\kappa_0)$.
5. $(u+v+w)^2 \le 3(u^2+v^2+w^2)$ and the fact that each block touches at most two coarse sites give $\sum_m \nu|\partial_m g|^2 \le 3B + (6g_7/(\varepsilon^2\kappa_0))A \le c_6(A+B)$.
6. Theorem 5.2: $\mathrm{Var}_\nu\, g \le (C_{\mathrm{site}}/(1-\rho_{\mathrm{col}}))\,c_6\,(\hbar/\varepsilon)(A+B)$.
7. Adding 2. and 6. gives the claim; (2) gives the numerical form, and $\ln(c_6 e^{80.79}) = 86.91$. □

Theorem 5.3 uses neither Theorem 3.1 with $L = 15$ nor Theorem 4.5 as stated; it uses their ingredients (path coupling, Lemmas 4.1, 4.3, 4.4). The factor ρ_{col} enters only through $1/(1-\rho_{\mathrm{col}})$; its numerical role is that of a single hypothesis, examined next.

6 The Dobrushin certificate

6.1 Why a certificate

A paper bound for ρ via a sup of $\partial_s \mathbb{E}[z_1 \mid z_8 = s]$ times $\mathrm{Var}(z_8)/(\varepsilon\hbar)$ does not close at small $\hbar$, and the curvature route cannot close it in principle:

Proposition 6.1 (curvature floor)**.** *For all boundary values with both neighbours of the middle site in the box and all $s \in [-x_c, 0]$, $F_\hbar''(s) \le 2/\varepsilon + \varepsilon \max V'' = 4.3441$. Hence every bound of the form $\mathrm{Var}_m \le \hbar/\kappa$ with $\kappa \le \inf F_\hbar''$ gives $\mathrm{Var}_m/(\varepsilon\hbar) \ge 0.4604$.*

Proof. $F_\hbar'' = \mathbb{E}[\partial_s^2 S \mid s] - \mathrm{Var}(\partial_s S \mid s)/\hbar \le 2/\varepsilon + \varepsilon V''(s)$, and $\max_{[-x_c,0]} V'' = V''(-x_c) = 0.68815$. □

At small $\hbar$ the middle site is pressed against the wall $-x_c$ by the slope $\varepsilon V'(-x_c) = 0.0463$ and its variance is of order $\hbar^2$, a boundary effect that curvature bounds do not see. The certificate below follows the law of the chain step by step instead.

6.2 Two lemmas

Let $\lambda = 1/(\varepsilon\hbar)$, $K(x) = e^{-\lambda x^2/2}$ and $[lo, hi] = [-x_c, 0]$. Define backward functions $g_N(v) = e^{-\varepsilon V(v)/\hbar} K(b - v)$, $g_i(v) = e^{-\varepsilon V(v)/\hbar} \int K(w - v) g_{i+1}(w)\, dw$.

Lemma 6.2 (influence recursion; Lemma R)**.** *Given $z_0 = a$, the chain $(z_1, \dots, z_N)$ is Markov with kernels $P_i(u, dv) = K(v - u) g_i(v) dv / \int K(v' - u) g_i(v') dv'$ (Doob h-transform). Put $f_8 = \mathrm{id}$, $f_{i-1} = P_i f_i$; then $\mathbb{E}[z_8 \mid a] = f_0(a)$, $\rho = f_0'(a)$, and*

$$f_{i-1}'(u) = \lambda \int f_i'(t)\, k_i(t; u)\, dt, \qquad k(t) := \mathrm{Cov}(V, \mathbf{1}\{V > t\}) = \bar F(t) \int_{lo}^{t} F + F(t) \int_{t}^{hi} \bar F \;\ge 0,$$

where $V \sim P_i(u, \cdot)$ with distribution function F and $\bar F = 1 - F$.

Proof. $\partial_u (P_i f)(u) = \lambda \,\mathrm{Cov}_{P_i(u,\cdot)}(V, f(V))$ because $\partial_u \log K(v - u) = \lambda(v - u)$; and $f(V) = f(lo) + \int f'(t) \mathbf{1}\{V > t\} dt$ gives $\mathrm{Cov}(V, f(V)) = \int f'(t) k(t) dt$ (Hoeffding [Hoe]). □

By induction $f_i' \ge 0$, without appeal to FKG. If $f_i' \le \Phi_i$ then $f_{i-1}' \le \lambda \int \Phi_i k_i$, and an upper bound for k follows from two-sided bounds $F_{lo} \le F \le F_{hi}$ alone, as a sum of products of nonnegative terms with no cancellation. Explicitly, for $t \in [t_j, t_{j+1}]$,

$$k(t) \le \big(1 - F_{lo}(t_j)\big) \int_{lo}^{t_{j+1}} F_{hi} + F_{hi}(t_{j+1}) \int_{t_j}^{hi} (1 - F_{lo}),$$

and the lower bound has the same form with the roles of F_{lo} and F_{hi} exchanged. For $V \equiv 0$ and an unbounded box the product of the eight factors is $\prod_{i=1}^{8} (16 - i)/(17 - i) = 1/2$, the exact value for the Gaussian bridge.

Lemma 6.3 (sandwiching; Lemma M)**.** (a) *$P_i(u, \cdot)$ is increasing in u in the likelihood-ratio order.* (b) *$g_i^{(b)}(v)$ is TP$_2$ in (v, b), so $P_i^{(b)}(u, \cdot)$ is increasing in b; the limit $b \to -\infty$ is the chain with $z_N = lo$.* (c) *Write $P_i(u, dw) \propto \exp(\lambda u w - \lambda w^2 + \chi_i(w))$. If on each cell of a partition $\chi_i' \in [s_j^-, s_j^+]$, then replacing χ_i by the piecewise linear functions with slopes $s_j^\pm$ (jumping only upwards, resp. downwards, at nodes) gives laws $L^- \le P_i(u, \cdot) \le L^+$ in the likelihood-ratio order.*

Proof. (a) The density is $e^{\lambda u v}$ times a function of v. (b) g_N is $e^{\lambda b v}$ times functions of v and of b; $K(w - v)$ is TP$_2$; compositions of TP$_2$ kernels are TP$_2$ by the 2×2 Cauchy–Binet formula [Ka]. (c) The log-ratio of the densities is monotone on each cell and across nodes. □

The Gaussian factor $e^{\lambda u w - \lambda w^2}$ times an exponential of a linear function is integrated exactly on each cell by the error function, so the strong push towards the wall at small $\hbar$ costs nothing in cell width; the remaining coarseness is the slope range $s_j^+ - s_j^-$.

The certificate inequality. On a cell $[u_l, u_{l+1}]$ of a and a cell $[b_{lo}, b_{hi}]$ of b one has $F_{L_i^+(u_{l+1})} \le F_{P_i} \le F_{L_i^-(u_l)}$, with cell masses given by closed erf formulas. Inserting these into Lemma 6.2 and running Φ_i backward as cellwise constants yields

$$\sup_{a\in[-x_c,0],\, b\in[b_{lo},b_{hi}]} \rho(a,b) \le \max_l \Phi_0(l).$$

The variable a is covered by cells continuously and b by the endpoint slopes of its cells; no reading between grid points of a or b remains. Reversing the roles gives a lower bound, used only as a check.

6.3 Computational discipline

The certificate (version 2) is an interval-arithmetic computation: every quantity that enters an inequality is carried as an interval with outward rounding, and the bound reported is the upper endpoint. The transcendental functions are not taken from the C library: exp is evaluated from a table together with a Taylor polynomial and an explicit remainder; erfcx from a table built with the interval arithmetic of `mpmath`, complete monotonicity between the nodes, and the asymptotic series with the remainder bound of [DLMF, §7.12(i)]; sums of floating-point numbers carry the rounding bound of [Hi]. The remaining arithmetic is IEEE double precision with round-to-nearest, widened outward after every operation. The certificate is written in the $\hbar$-interval form (slopes written about the centre of each law), so that each of the 239 intervals of $\hbar$ is covered as a whole and not at sample points. The assumptions are: (1) IEEE 754 double precision with round-to-nearest for the basic operations, and the correctness of the outward widening (`nextafter`) and of the summation bound as implemented; (2) the correctness of the interval arithmetic of `mpmath`; (3) the remainder bound [DLMF, §7.12(i)]; (4) the summation bound [Hi]; (5) the box endpoint x_c is read as the double-precision number $\mathrm{fl}(0.89)$. Negative controls: versions of the program with one component disabled (no outward widening; exp without remainder; erfcx without remainder) fail an enclosure check against 60-digit and exact rational evaluations in 147958, 7713 and 3669 cases respectively, while the program as used fails in none. Version 1 of this note used double precision with outward margins (not interval arithmetic) at 23 values of $\hbar$. The same double-precision method, run on the 239 intervals, gives bounds that agree with the interval bounds to relative $4\cdot 10^{-4}$ for $\hbar \ge 10^{-4}$ and are larger below (by up to 20% on $[10^{-5}, 3\cdot 10^{-5}]$ and by a factor of up to 3.4 on seven intervals near $3\cdot 10^{-5}$, where its margins on the end geometries were far more conservative); it is never smaller. Those values are not used.

6.4 The interval certificate on $[10^{-5}, 1]$ and the large-$\hbar$ range

Proposition 6.4 (DOB)**.** (a) [paper, plus one floating-point value] *Under the convention that for $k \equiv 0 \pmod 8$ the coarse sites are $y_m = z_{8m-1}$ (so that both neighbours of every coarse site lie in the box), for $\hbar \in [0.27, 1]$ one has $\sup\rho \le (V_{\mathrm{in}}(0.27)/(0.27\,\varepsilon))^2$ and hence $\rho_{\mathrm{col}} \le 0.858$, provided $V_{\mathrm{in}}(0.27) \le 0.0884$. Here $V_{\mathrm{in}}(\hbar) = \min_\theta \max(\mathbb{E}_{\nu_R}(X-\theta)^2, \mathbb{E}_{\nu_L}(X-\theta)^2)$ with $\nu_R \propto e^{-(x^2/\varepsilon+\varepsilon V(x))/\hbar}$ on $[\theta, 0]$ and $\nu_L \propto e^{-((x+x_c)^2/\varepsilon+\varepsilon V(x))/\hbar}$ on $[-x_c, \theta]$. The value $V_{\mathrm{in}}(0.27) = 0.08744$ was computed in floating point by the trapezoidal rule, without interval arithmetic or outward margins.*

(b) [interval arithmetic, under assumptions (1)–(5) of §6.3] *For every $\hbar \in [10^{-5}, 1]$, for every geometry of the coarse chain (interior $r = 7$, $b \in [-x_c, 0]$; ends $r = 0, \dots, 7$, $b \in (-\infty, -x_c]$) and every $a \in [-x_c, 0]$, $\sup\rho \le 0.1997$; hence $\rho_{\mathrm{col}} \le 0.400 < 1$. The range $[10^{-5}, 1]$ is covered by 239 intervals of $\hbar$ without gaps (Table 1), each certified over the whole interval.*

(c) *For $0 < \hbar \le 10^{-5}$ see Proposition 6.6.*

Proof of (a). For a positive density p on $[l,r]$ with $(\log p)' \le c_R$ on (θ,r) and $(\log p)' \ge -c_L$ on (l,θ), $\mathbb{E}_p(X-\theta)^2 \le \max(\mathbb{E}_{\nu_R}(X-\theta)^2, \mathbb{E}_{\nu_L}(X-\theta)^2)$, by the Chebyshev order inequality on each side (the ratio p/ν decreases in $|X-\theta|$ while $(X-\theta)^2$ increases). For the middle site, $-\partial_s S \le -2s/\varepsilon - \varepsilon V'(s)$ (both neighbours at 0) and $-\partial_s S \ge -[2(s+x_c)/\varepsilon + \varepsilon V'(s)]$ (both at the wall), which are the densities ν_R, ν_L; the same bounds hold for z_1 and z_7 in the left block. With the Hoeffding representation $\rho = (\varepsilon\hbar)^{-1}\int \varphi'(t)H(t)\,dt$, $\varphi(s) = \mathbb{E}[z_1 \mid z_8 = s]$, $H \ge 0$, $\int H = \mathrm{Var}_m$ [Hoe], and $\varphi' = \mathrm{Cov}(z_1, z_7 \mid s)/(\varepsilon\hbar) \le V_{\mathrm{in}}/(\varepsilon\hbar)$ by Cauchy–Schwarz, one gets $\sup\rho \le (V_{\mathrm{in}}(\hbar)/(\varepsilon\hbar))^2$. The comparison densities are monotone in $\hbar$ in the likelihood-ratio order, so $V_{\mathrm{in}}(\hbar)/(\varepsilon\hbar)$ is non-increasing and the value at $\hbar = 0.27$ suffices; $2(0.0884/0.135)^2 < 0.86$ is checked in Lean. The use of the convention for $k \equiv 0 \pmod 8$ in Theorem 5.3 only needs block lengths ≤ 7 and at most 14 sites around a coarse site, which the shifted division keeps. □

Part (b) is the output of the certificate of §6 in its interval form; its proof is the computation, whose programs, inputs (the table of 239 intervals), outputs and hashes are distributed (§7). The computation took 33 CPU-hours on one machine; it was run as four processes on disjoint sets of intervals with identical settings and merged, and the merged output was checked independently (all 239 intervals present, no gap, one set of settings). Table 1 reports, for each range of $\hbar$, the number of intervals, the largest upper bound and the largest lower bound (a check, not used); on every interval lower $\le$ upper.

range of $\hbar$	intervals	c_1	upper (max)	lower (max)	CPU (s)
$[10^{-5}, 3\cdot10^{-5}]$	23	0.2	0.1979	$6.4\cdot10^{-7}$	57210
$[3\cdot10^{-5}, 10^{-4}]$	21	0.12	0.1989	$2.6\cdot10^{-5}$	17162
$[10^{-4}, 2\cdot10^{-4}]$	13	0.08–0.15	**0.1997**	$1.5\cdot10^{-4}$	7431
$[2\cdot10^{-4}, 4\cdot10^{-4}]$	10	0.06	0.1992	$2.9\cdot10^{-4}$	7026
$[4\cdot10^{-4}, 10^{-3}]$	15	0.05	0.1953	$9.9\cdot10^{-4}$	7370
$[10^{-3}, 10^{-2}]$	52	0.04	0.1923	$5.2\cdot10^{-3}$	13569
$[10^{-2}, 0.1]$	95	0.03	0.1828	$6.6\cdot10^{-3}$	9159
$[0.1, 1]$	10	0.03	0.1361	$5.9\cdot10^{-4}$	138

Table 1: The interval certificate of Proposition 6.4(b): the 239 intervals of $\hbar$, grouped by range. c_1 is the width parameter of the u-cells (width about $c_1\sqrt{\varepsilon\hbar}$). "Upper" is the interval-arithmetic bound on $\sup\rho$ over all geometries, maximised over the intervals of the range; "lower" is the interval lower bound (a check). The worst interval is $[1.325\cdot10^{-4}, 1.433\cdot10^{-4}]$ (interior geometry, $b \in [-0.01, 0]$), with upper bound 0.1997.

Controls. With the division every fourth site (blocks of length 3) the certificate returns upper bounds 2.20, 10.6, 365 at $\hbar = 0.03, 0.01, 10^{-3}$, so it does not certify $\rho_{\mathrm{col}} < 1$ there (necessarily, being an upper bound; this is not a test of power). For the Gaussian chain ($V \equiv 0$), whose exact value is $1/2$, it returns the enclosure $[0.061, 3.63]$, which contains $1/2$ and is loose.

Corollary 6.5. *Under the hypotheses of Theorem 5.3 other than* (DOB)*: for $\hbar \in [0.27, 1]$, with the qualification of Proposition 6.4*(a)*, $C_P(Q_k) \le 9.18 + e^{88.87}\hbar^{-8}$ for all k; for every $\hbar \in [10^{-5}, 1]$, under the assumptions of Proposition 6.4*(b)*, $C_P(Q_k) \le 9.18 + e^{87.43}\hbar^{-8}$ for all k; and for every $\hbar \in (0, 10^{-5}]$, by Proposition 6.6, $C_P(Q_k) \le 9.18 + e^{88.99}\hbar^{-8}$ for all k. In particular $C_P(Q_k) \le 9.18 + e^{88.99}\hbar^{-8}$ for all $\hbar \in (0, 1]$ and all k.*

6.5 Geometries and cells

The supremum defining ρ_{col} runs over every conditional law that occurs in the coarse chain. We cover them as follows. Interior coarse sites ($r = 7$): $b \in [-x_c, 0]$ is split into the nine cells with endpoints $-0.89, -0.6, -0.4, -0.25, -0.15, -0.08, -0.04, -0.02, -0.01, 0$. Coarse sites next to the end of the chain ($r = 0, \dots, 7$): $b = \beta \in (-\infty, -x_c]$ is split into cells whose lowest member is handled by the limit $b \to -\infty$ of Lemma 6.3(b); for $r \le 2$ the cells near $-x_c$ have width of order $\sqrt{\varepsilon\hbar}$. The first coarse site is the mirror image of the last. The variable $a \in [-x_c, 0]$ is covered by the u-cells of width about $c_1\sqrt{\varepsilon\hbar}$, with $c_1 \in [0.015, 0.25]$ depending on $\hbar$; the column "cells" of Table 1 is their number.

At $\hbar = 10^{-5}$ the end geometries with $r \le 2$ need narrow b-cells: for $r = 0$ the centre $(u + b)/2$ of the last kernel moves by half the width of the b-cell, and a cell much wider than $\sqrt{\varepsilon\hbar}$ makes the sandwich of Lemma 6.3(c) much wider than the law itself. In the interval certificate the width parameter c_1 of the u-cells is 0.03 for $\hbar \ge 10^{-2}$ and increases to 0.2 for $\hbar \le 3 \cdot 10^{-5}$ (Table 1), following the table of intervals fixed in advance; the smallest range of $\hbar$ accounts for about half of the computing time. This affects only the cost of the computation, not its validity: Lemma 6.3(c) holds for every partition.

6.6 Small $\hbar$: pinning to the wall

Proposition 6.6 (PIN)**.** [paper, plus one one-dimensional interval computation; algebra in Lean] *For every $0 < \hbar \le 10^{-5}$ and every geometry of the coarse chain,*

$$0 \le \rho(a, b) \le \frac{2L_g\,\varepsilon\hbar}{d(\hbar)^2} + \mathrm{tail} \le 0.437, \qquad \textit{hence } \rho_{\mathrm{col}} \le 0.874 < 1,$$

where $L_g = 1980/1217$, $d(\hbar) = D_0 - 2\sqrt{7h_{77}\varepsilon\hbar}$, $h_{77} = 3664648/1770735$, $D_0 = 0.023126$ is an interval-arithmetic lower bound for the classical force at the wall in the worst geometry, and the tail is at most $x_c^3\lambda M e^{-\lambda\Delta}$ with $\lambda = 1/(\varepsilon\hbar)$, $M = 2$, $\Delta \ge 0.0041$ (numerically zero). The bound is increasing in $\hbar$, so its value at $\hbar = 10^{-5}$ is the supremum, and $\rho = O(\hbar)$ as $\hbar \to 0$.

Proof sketch. (L0) A bridge of 7 sites between fixed endpoints has Hessian at least $H_0 = \mathrm{tridiag}(-1, 15/8, -1)$ (since $\varepsilon^2 V'' \ge -1/8$), which is positive definite, while for 8 sites $\det H_0^{(8)} < 0$: seven is within the conjugate length, eight is not. (L1) Given z_8, z_1 and the rest are conditionally independent of a except through the left bridge, so $\rho = \lambda\,\mathrm{Cov}(z_8, g(z_8))$ with $g(u) = \mathbb{E}[z_1 \mid z_8 = u]$. (L2) On the convex left bridge the Brascamp–Lieb inequality [BL] and FKG give $0 \le g' \le L_g = (h_{11} + h_{77} + 2h_{17})/4$ with h_{ij} the entries of $(H_0^{(7)})^{-1}$; hence $\rho \le \lambda L_g\,\mathrm{Var}(z_8) \le \lambda L_g\,\mathbb{E}[(z_8 + x_c)^2]$. (L3) Integration by parts on the convex box gives $\mathbb{E}[z_7] \le m_7 + \sqrt{7h_{77}/\lambda}$, where m_7 is the value at site 7 of the classical minimiser of the bridge. (L4) The marginal of z_8 has density proportional to $e^{-\lambda W}$ with $W'(u) = 2u - \mathbb{E}_L[z_7] - \mathbb{E}_R[z_9] + \varepsilon^2 V'(u)$. (L5) By Topkis' comparison [To] (the coupling $-z_i z_{i+1}$ is submodular and the bridges are strictly convex) the classical minimisers are monotone in the boundary values, so the worst geometry is $a = 0$ (and $b = 0$ in the interior, $b = -x_c$ at the ends): $W' \ge D(u) - 2e$ with $e = \sqrt{7h_{77}\varepsilon\hbar}$ and D the classical force in that geometry. On $[-x_c, -0.8]$, $D \ge D_0$ by a supersolution lemma (submodularity and convexity of the bridge) checked in interval arithmetic, so $W' \ge d := D_0 - 2e > 0$ and the law of z_8 on that range is stochastically dominated by an exponential law of rate λd, whence $\mathbb{E}[(z_8 + x_c)^2] \le 2/(\lambda d)^2$; on $[-0.8, 0]$ the mass is bounded by the tail stated. At $\hbar = 10^{-5}$: $e = 0.00851$, $d = 0.00610$, bound 0.437. The exact rational values $\det H_0^{(7)} > 0$, $\det H_0^{(8)} < 0$, h_{11}, h_{17}, L_g, the assembly and the final inequality $2L_g\varepsilon\hbar/d^2 \le 0.437$ for $0 < \hbar \le 10^{-5}$ are in Lean (`LatticeGauge.PinnedCoarseChain`, §7); the analysis (L1)–(L5) is on paper. The complete argument, with the interval computation for D_0, is distributed with this note (`certificates/pin/`). □

The bound of Proposition 6.6 is about three orders of magnitude above the measured influence at $\hbar = 10^{-5}$ (which is $1.4 \cdot 10^{-4}$); the loss comes from the $\sqrt{n}$ of the integration by parts and from not using the pinning of z_7 itself. At the junction $\hbar = 10^{-5}$ the two bounds are 0.874 (PIN) and 0.400 (certificate).

6.7 Where the result sits in the conditional argument

In the conditional argument for the double-well path measure in [KSR, Part 10], the Poincaré constant $C_{(\mathrm{T0}'_P)}(\hbar) := \sup_{k,\alpha,\beta} C_P(Q_k)$ enters a second-order term of the form $C_{(\mathrm{T0}'_P)} \cdot D/\xi^2$, where ξ is the tunnelling correlation length, which grows like $e^{c/\hbar}$. Any bound polynomial in $\hbar$ therefore makes that term vanish as $\hbar \to 0$, however crude its constant. Theorem 5.3 with Propositions 6.4 and 6.6 provides such a bound for every $\hbar \in (0, 1]$, under the qualifications stated there. In [KSR], Proposition 4.5 is the first step of this argument for the $0 + 1$-dimensional double well (Parts 10-05 and 10-16); its provisional leaves are the seven rows [01]–[07], of which ($\mathrm{T0}'_P$) is [04] and the cross-term row (X), which does not involve $C_{(\mathrm{T0}'_P)}$, is [06]. The present result bears on [04] only. Of the two ways in which [KSR, 10-12] supports ($\mathrm{T0}'_P$), the box-side core (rows [082] and [084]) and the compartment-side core (rows [099] and [018]), it bears on the second, whose core has both outer values in the well; the other six leaves, including (X), are unchanged.

7 What is machine-checked

The following Lean 4 modules (Mathlib [M]) contain the algebraic and finite steps; measure, integration, limits and the analytic lemmas are outside Lean.

- `LatticeGauge.HiddenConvexity`: `geom_mean_midpoint`, `act_eq_actRho`, `actRho_midpoint`, `act_min_unique` (K′ for $a, b \le 0$, midpoint form).
- `LatticeGauge.DoeblinPoincare`: `path_coupling_step`, `poincare_of_block_contraction`, `spectral_weight_zero`, `contraction_rate_pos` (Theorem 3.1).
- `LatticeGauge.BlockPoincare`: `kappa_seven_gt` ($\kappa_0 > 1/20$), `free_energy_quasiconvex`, `quasiconvex_add_osc`, `ratio_exp_bound`, `hardy_discrete`, `two_scale_assembly` (Lemmas 4.2–4.4).
- `LatticeGauge.CoarseChainPoincare`: `cov_le_lip_var`, `cross_term`, `sq_three`, `coarse_grad_sum`, `path_coupling_site`, `poincare_of_site_contraction`, `two_level_assembly`, `coarse_chain_poincare`, `g7_value`, `assembly_const` (Theorems 5.2, 5.3).
- `LatticeGauge.InfluenceRecursion`: `tp2_comp`, `cdf_upper`, `cdf_lower`, `k_upper`, `step_bound`, `gauss_product` (Lemmas 6.2, 6.3).
- `LatticeGauge.CenterCovariance`: `hoeffding_kernel`, `kernel_le_sup`, `chebyshev_double`, `large_hbar_column` (Proposition 6.4(a)).
- `LatticeGauge.DobrushinIntervalTable` (version 2): `centered_split`, `geomean_dev`, `mul_le_corners`, `steps_growth`, `geometric_cover_count`, `cover_le`, `window_le_of_interval_table`, `column_lt_one_of_table`, `join_windows` (the passage from the table of $\hbar$-intervals to the windows of Proposition 6.4(b); the interval arithmetic itself is outside Lean).

- `LatticeGauge.PinnedCoarseChain` (version 2): `D_six`, `D_seven`, `D_pos_le_seven`, `D_eight_neg`, `h11_value`, `h17_value`, `Lg_value`, `assemble`, `final_bound` (Proposition 6.6: the exact rational constants and the final inequality).

Each statement carries its hypotheses in the Lean signature. The independent re-check of the seven modules of version 1 was completed on 2026-10-15 (every file checked with `lake env lean` in the same environment on a separate session, rc=0, standard axioms only); the two modules of version 2 were checked in the same way on 2026-10-17, and the bundle and the axioms file at the addresses below were updated on that day. The bundle is listed at `https://computoergosum.com/principia/lean/principia-ym-coarse-chain-poincare-2026-10-15.tar.gz` (with a `.sha256` file at the same address) and the complete `#print axioms` output at `https://computoergosum.com/principia/lean/axioms-ym-coarse-chain-poincare-2026-10-15.txt`; both addresses were placed on 2026-10-15, after the independent re-check. To check: unpack, run `lake exe cache get`, then `LEAN_NUM_THREADS=1 lake build` on each module above, and compare the trailing `#print axioms` lines with the log (expected: `propext`, `Classical.choice`, `Quot.sound` only). The ledger and build instructions of the site's Lean developments are at `https://computoergosum.com/en/principia/lean/index.html`. The certificate's programs, specification files and output files (one JSON per run, with every cell's upper and lower value), a manifest of the files behind Table 1 (`certificates/interval/`, with the table of 239 intervals, the merged output of the four processes and their logs; the files of version 1 remain in `certificates/`), the record and the one-dimensional interval computation of Proposition 6.6 (`certificates/pin/`), and SHA-256 hashes are part of the same bundle.

8 Correspondence table

Line numbers refer to the files `LatticeGauge<Module>.lean` of the bundle; the third column gives `<Module>`: line of each declaration in the order of the second column.

paper	Lean declaration (namespace `LatticeGauge.`) and what is outside	file: line
(K′), §2	`actRho_midpoint`, `act_min_unique`; partial minimisation on paper	`HiddenConvexity`: 75, 115
Theorem 3.1	`path_coupling_step`, `spectral_weight_zero`, `poincare_of_block_contraction`; FKG, Strassen, spectral theorem on paper	`DoeblinPoincare`: 16, 54, 29
Lemma 4.1	`kappa_seven_gt`; Bakry–Émery cited	`BlockPoincare`: 26
Lemma 4.2	`two_scale_assembly`, `cross_term_sq`, `deriv_sq_split`; total variance and differentiation on paper	`BlockPoincare`: 101, 88, 96
Lemma 4.3	`hardy_discrete`, `ratio_exp_bound`; Fubini on paper	`BlockPoincare`: 137, 66
Lemma 4.4	`free_energy_quasiconvex`, `quasiconvex_add_osc`; Laplace-type bounds on paper	`BlockPoincare`: 37, 51
Lemma 5.1, Lemma 6.3(b)	`tp2_comp` (finite form)	`InfluenceRecursion`: 18
Theorem 5.2	`path_coupling_site`, `poincare_of_site_contraction`	`CoarseChainPoincare`: 92, 98
Theorem 5.3	`cross_term`, `sq_three`, `coarse_grad_sum`, `two_level_assembly`, `coarse_chain_poincare`, `g7_value`, `g7_lt`, `assembly_const`; BL cited	`CoarseChainPoincare`: 51, 47, 68, 78, 117, 128, 133, 136
Lemma 6.2	`cdf_upper`, `cdf_lower`, `k_upper`, `step_bound`, `gauss_product`	`InfluenceRecursion`: 49, 56, 60, 68, 76
Proposition 6.1	paper only	—
Proposition 6.4(a)	`hoeffding_kernel`, `kernel_le_sup`, `chebyshev_double`, `large_hbar_column`; the value $V_{\mathrm{in}}(0.27)$ outside	`CenterCovariance`: 44, 143, 186, 206
Proposition 6.4(b)	interval-arithmetic certificate outside Lean; the passage from the table of $\hbar$-intervals to the windows: `geometric_cover_count`, `cover_le`, `window_le_of_interval_table`, `column_lt_one_of_table`, `join_windows`	`DobrushinIntervalTable` (version 2)
Proposition 6.6	`D_seven`, `D_eight_neg`, `h11_value`, `h17_value`, `Lg_value`, `assemble`, `final_bound`; analysis on paper, BL and Topkis cited	`PinnedCoarseChain` (version 2)

9 What is not claimed

1. **Assumptions of the certificate.** Proposition 6.4(b) rests on assumptions (1)–(5) of §6.3. The interval arithmetic itself is not formalised; the Lean module covers only the passage from the table of intervals to the windows.

2. **The junction at $\hbar = 10^{-5}$ and the small-$\hbar$ constant.** Proposition 6.6 gives $\rho_{\mathrm{col}} \le 0.874$, against 0.400 from the certificate on $[10^{-5}, 1]$; its bound is about three orders of magnitude above the measured influence. Its cited tools (Brascamp–Lieb on convex sets, Topkis' comparison) and the integration by parts are not formalised.

3. **Cited statements.** BE_7 (Bakry–Émery on convex domains), BL (Brascamp–Lieb on convex domains) and the FKG/Holley step of Theorem 3.1 are standard results used as cited; they are not formalised in Lean.

4. **Constants.** $e^{87}\hbar^{-8}$ is crude; measured block and site constants are of order 10^2 and below 0.27 respectively. Only a Poincaré inequality is proved, not a log-Sobolev inequality.

5. **One dimension.** The model is a one-dimensional chain (a 0+1-dimensional path measure); nothing is claimed about four-dimensional lattice gauge theory. In the conditional argument of [KSR, Part 10] for this double well, Proposition 4.5 keeps the seven provisional leaves

row	content	status
END	outer boundary values $\alpha, \beta \le -x_c$	hypothesis of every statement
K′	convexity of $\tilde{S}$ in $\rho = z^2$, jointly with (a^2, b^2)	paper; algebraic core in Lean
BE_7	$C_P \le \varepsilon/\kappa(n)$ for blocks of length ≤ 7	paper ($\kappa_0 > 1/20$ in Lean) + standard, cited [BoL, Prop. 3.1], [BGL], [BL]
BL	$\mathrm{Var}\langle e, z\rangle \le \hbar\langle e, M^{-1}e\rangle$ on convex sets	standard, cited [BL]; g_7 in Lean
TP_2	monotonicity of the coarse conditional laws	paper [Ka]; finite form in Lean
FKG	monotone coupling for blocks (Theorem 3.1 only)	standard, cited [Ho], [Pr]
H1D, QC	Hardy-type bound; quasi-convexity plus oscillation	paper; discrete cores in Lean
OLL	contraction implies gap	cited, checked (arXiv version) [Ol]
ALG	coupling step, comparison, assembly, $c_6 < 497$	Lean
DOB, $[0.27, 1]$	$\rho_{\mathrm{col}} \le 0.858$	paper + one floating-point value
DOB, $\hbar \in [10^{-5}, 1]$	$\rho_{\mathrm{col}} \le 0.400$ (239 intervals, no gap)	interval arithmetic, under assumptions (1)–(5) of §6.3; table-to-window algebra in Lean
PIN, $\hbar \in (0, 10^{-5}]$	$\rho_{\mathrm{col}} \le 0.874$	paper; BL on convex sets [BL] and Topkis [To] cited; D_0 by one-dimensional interval arithmetic; algebra in Lean

Table 2: Status of the rows on which Theorem 5.3 and Corollary 6.5 rest.

[01]–[07]; this paper bears only on [04] = ($\mathrm{T0}'_P$), through its compartment-side support, and leaves the other six, including the cross-term row (X) = [06], unchanged.

6. **No external review.** The proofs, the computations and this text were produced within the authors' group (§10).

10 Disclosure of AI use

This note was produced with the assistance of AI agents. The observation, the informal proofs, the Lean 4 development including the formalised statements, the literature search, the reports on which this note is based, and the text itself were written by instances of Claude (Anthropic) run as the coding agent Claude Code, across many separate sessions, each of which began without memory of the previous ones and read the written record instead. A second group of instances, running on the server of the website named on the title page, is responsible for rebuilding the Lean development from the distributed archive on a separate machine, for checking the claims of this note against the reports, and for preparing the distributed version; the independent rebuild was completed on 2026-10-15 (every file checked with `lake env lean` in the same environment on a separate session, rc=0, standard axioms only). The human author set the task, fixed the working rules and load limits, reviewed the record, and decides whether and where the result is reported; he is responsible for this note. Adversarial review was carried out only by further instances of the same model family, which share the same blind spots. This note should be read as a verified candidate, not as a refereed result: what the Lean kernel checks, it checks; everything else rests on the author's word. In particular the analytic steps listed in §7 are checked on paper only. This edition and the Lean development it cites are kept in the repository `https://github.com/yoshikazuhosomura-sys/principia` (the note under `papers/ym-coarse-chain-poincare`, the development under `lean/ym-coarse-chain-poincare`). The web edition of this note,

published on the site named on the title page, lists the same work under the names Kiichi, Shiori and Rin, the human author and the two groups of AI agents; the two editions differ only in the title page, in this section and in the naming of the agent that carried out the independent re-check.

Version 2 (2026-10-17) adds the interval-arithmetic certificate over $[10^{-5}, 1]$ (Proposition 6.4(b)) and Proposition 6.6. The certificate was run on one machine (33 CPU-hours, four processes on disjoint sets of intervals, merged); its merged output, the Lean modules of version 2 and the arithmetic of the constants were checked by a second instance of the agent, independent of the one that produced them.

References

[BGL] D. Bakry, I. Gentil and M. Ledoux, *Analysis and Geometry of Markov Diffusion Operators*, Grundlehren Math. Wiss. 348, Springer, 2014, DOI 10.1007/978-3-319-00227-9; cited for the curvature criterion on convex domains (not consulted directly for this statement).

[BL] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation*, J. Funct. Anal. 22 (1976), 366–389, DOI 10.1016/0022-1236(76)90004-5; cited for Thm. 4.1 (not consulted directly).

[BoL] S. G. Bobkov and M. Ledoux, *From Brunn–Minkowski to Brascamp–Lieb and to logarithmic Sobolev inequalities*, Geom. Funct. Anal. 10 (2000), 1028–1052; cited for Prop. 3.1 (the curvature criterion on convex domains).

[DLMF] NIST Digital Library of Mathematical Functions, §7.12(i), `https://dlmf.nist.gov/7.12`; cited for the remainder bound of the asymptotic expansion of erfc.

[Do] R. L. Dobrushin, *Prescribing a system of random variables by conditional distributions*, Theory Probab. Appl. 15 (1970), 458–486, DOI 10.1137/1115049 (not consulted directly).

[Hi] N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, 2002; cited for the standard bound on the rounding error of a floating-point sum (referred to as (5.3) in the documentation of the certificate; not consulted directly).

[Hoe] W. Hoeffding, *Masstabinvariante Korrelationstheorie*, Schr. Math. Inst. Univ. Berlin 5 (1940), 181–233 (not consulted directly).

[Ho] R. Holley, *Remarks on the FKG inequalities*, Comm. Math. Phys. 36 (1974), 227–231, DOI 10.1007/BF01645980 (not consulted directly).

[HS] R. Holley and D. Stroock, *Logarithmic Sobolev inequalities and stochastic Ising models*, J. Stat. Phys. 46 (1987), 1159–1194, DOI 10.1007/BF01011161 (not consulted directly).

[Ka] S. Karlin, *Total Positivity*, Vol. I, Stanford University Press, 1968 (not consulted directly).

[KSR] Kiichi, Shiori and Rin, *Yang–Mills and the mass gap* (online textbook, in Japanese), Part 10, chapters on the double-well path measure and on computer-assisted proofs, `https://computoergosum.com/principia/yang-mills/index.html` (2026).

[KSR2] Kiichi, Shiori and Rin, *The inverse squared spectral gap bounds the Langevin Poincaré constant of Euclidean path measures from below*, 2026-10-09, `https://computoergosum.com/en/index.html`.

[Ma] F. Martinelli, *Lectures on Glauber dynamics for discrete spin models*, in: Lectures on Probability Theory and Statistics (Saint-Flour 1997), Lecture Notes in Math. 1717, Springer, 1999, 93–191, DOI 10.1007/978-3-540-48115-7_2.

[M] Mathlib, `https://github.com/leanprover-community/mathlib4`.

[Mu] B. Muckenhoupt, *Hardy's inequality with weights*, Studia Math. 44 (1972), 31–38 (not consulted directly).

[Ol] Y. Ollivier, *Ricci curvature of Markov chains on metric spaces*, J. Funct. Anal. 256 (2009), 810–864, DOI 10.1016/j.jfa.2008.11.001, arXiv:math/0701886; cited (arXiv v4) for Prop. 29 and Cor. 30.

[Pr] C. J. Preston, *A generalization of the FKG inequalities*, Comm. Math. Phys. 36 (1974), 233–241, DOI 10.1007/BF01645981 (not consulted directly).

[SZ] D. W. Stroock and B. Zegarlinski, *The logarithmic Sobolev inequality for discrete spin systems on a lattice*, Comm. Math. Phys. 149 (1992), 175–193, DOI 10.1007/BF02096629.

[To] D. M. Topkis, *Supermodularity and Complementarity*, Princeton University Press, 1998; cited for the monotone comparison of minimisers (not consulted directly).